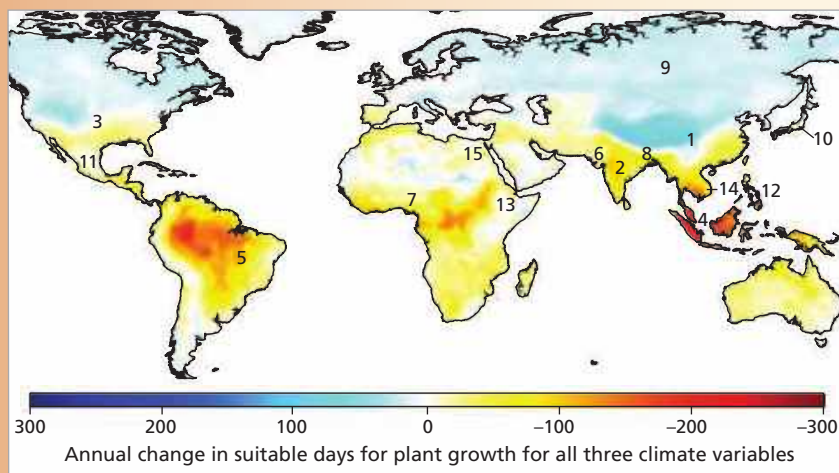


## PROBLEM-SOLVING EXERCISE

### How will climate change affect crop productivity?

Plant growth is significantly limited by air temperature, water availability, and solar radiation. A useful parameter for estimating crop productivity is the number of days per year when these three climate variables are suitable for plant growth. Camilo Mora (University of Hawaii at Manoa) and his colleagues analyzed global climate models to project the effect of climate change on suitable days for plant growth by the year 2100.



**Map data from** Camilo Mora, et al. Days for Plant Growth Disappear under Projected Climate Change: Potential Human and Biotic Vulnerability. *PLoS Biol* 13(6): e1002167 (2015).

→ **Instructors:** A version of this Problem-Solving Exercise can be assigned in **Mastering Biology**.

In this exercise, you will examine projected effects of climate change on crop productivity and identify the resulting human impacts.

**Your Approach** Analyze the map and table. Then answer the questions below.

**Your Data** The researchers projected the annual changes in suitable days for plant growth for all three climate variables: temperature, water availability, and solar radiation. They did so by subtracting recent averages (1996–2005) from projected future averages (2091–2100). The map shows the projected changes if no measures are taken to reduce climate change. The numbers identify locations of the 15 most populous nations. The table identifies their economies as either mainly industrial (factory icon) or agricultural (plant icon) and their annual per capita income category.

| Nation        | Map location | Estimated population in 2014 (millions) | Type of economy | Income category* |
|---------------|--------------|-----------------------------------------|-----------------|------------------|
| China         | 1            | 1,350                                   | Industrial      | \$\$\$           |
| India         | 2            | 1,221                                   | Agricultural    | \$\$             |
| United States | 3            | 317                                     | Industrial      | \$\$\$\$         |
| Indonesia     | 4            | 251                                     | Agricultural    | \$               |
| Brazil        | 5            | 201                                     | Agricultural    | \$\$\$           |
| Pakistan      | 6            | 193                                     | Agricultural    | \$\$             |
| Nigeria       | 7            | 175                                     | Agricultural    | \$\$             |
| Bangladesh    | 8            | 164                                     | Agricultural    | \$               |
| Russia        | 9            | 143                                     | Industrial      | \$\$\$\$         |
| Japan         | 10           | 127                                     | Industrial      | \$\$\$\$         |
| Mexico        | 11           | 116                                     | Agricultural    | \$\$\$           |
| Philippines   | 12           | 106                                     | Agricultural    | \$\$             |
| Ethiopia      | 13           | 94                                      | Agricultural    | \$               |
| Vietnam       | 14           | 92                                      | Agricultural    | \$               |
| Egypt         | 15           | 85                                      | Agricultural    | \$\$             |

Data from The World Bank

\*Per capita income, based on World Bank categories: \$ = low: < \$1,035; \$\$ = lower middle: \$1,036–\$4,085; \$\$\$ = upper middle: \$4,086–\$12,615; \$\$\$\$ = high: > \$12,615.

### Your Analysis

1. Camilo Mora began the study as a result of talking with someone who claimed that climate change improves plant growth because it increases the number of days above freezing. Based on the map data, how would you respond to this claim?
2. What do the map and the table data indicate about the impact of the projected changes on humans?

### Drought

On a sunny, dry day, a plant may wilt because its water loss by transpiration exceeds water absorption from the soil. Prolonged drought, of course, will kill a plant, but plants have control systems that enable them to cope with less extreme water deficits.

Many of a plant's responses to water deficit help the plant conserve water by reducing the rate of transpiration. Water deficit in a leaf causes stomata to close, thereby slowing transpiration dramatically (see Figure 36.14). Water deficit stimulates increased synthesis and release of abscisic acid in the leaves; this hormone helps keep stomata closed by acting

on guard cell membranes. Leaves respond to water deficit in several other ways. For example, when the leaves of grasses wilt, they roll into a tubelike shape that reduces transpiration by exposing less leaf surface to dry air and wind. Other plants, such as ocotillo (see Figure 36.15), shed their leaves in response to seasonal drought. Although these leaf responses conserve water, they also reduce photosynthesis, which is one reason why a drought diminishes crop yield. Plants can even take advantage of early warnings in the form of chemical signals from wilting neighbors and prime themselves to respond more readily and intensely to impending drought stress (see the **Scientific Skills Exercise**).